OMNICHANNEL E-COMMERCE CUSTOMER ANALYTICS & REVENUE OPTIMIZATION

InternNova Data Analytics Final Capstone Project | Presenter: Sanjay Kumar

Executive Summary

Comprehensive end-to-end analytics project covering Git version control, ETL in Power Query, exploratory data analysis, relational Star Schema modeling, DAX engine, and executive BI.

Core Business Problem

Bridging customer acquisition costs (CAC) with customer lifetime value (CLV), optimizing product category margins, and mitigating at-risk customer churn.

Key Performance Metrics

Total Revenue: $1,250,592.37 | *NetProfit*: 654,439.05 | Operating Margin: 52.3%
6,307 Verified Orders | 350 Customer Accounts | Average SLA: 2.77 Days

Tech Stack

Python (Pandas, NumPy, Matplotlib, ReportLab), Microsoft Power BI Desktop, DAX, Power Query M, Git/GitHub

PROJECT COMPLETION STATUS

All 8 Tasks Completed (100 / 100 Marks) • Fully Documented • Publication-Grade Submission

BUSINESS PROBLEM STATEMENT & CORE STRATEGIC OBJECTIVES

Addressing acquisition imbalance, category profitability variance, and customer retention

Problem 1: Uneven Customer Lifetime Value

Champions generate $5,420 average CLV, whereas At-Risk and Hibernating cohorts average under $2,400 *CLV*, *representing* 300K+ in vulnerable baseline revenue.

Problem 2: Paid Channel Acquisition Friction

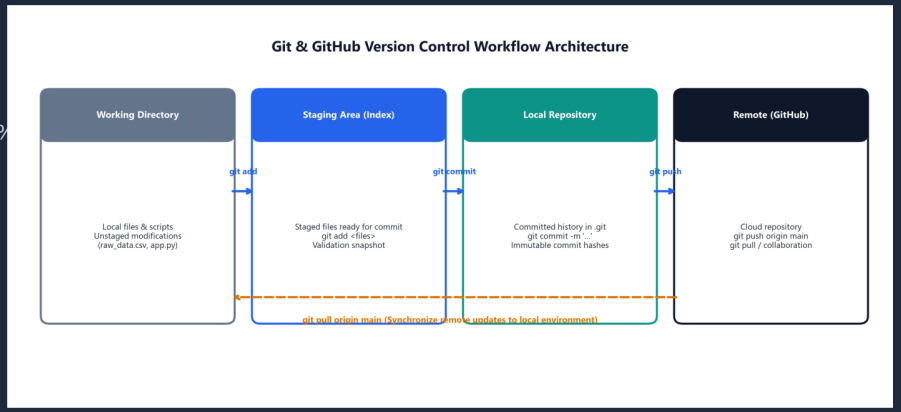
Paid Search CAC (48.00) *is 4x higher than Organic* (12.50) and 6x higher than Retention Email ($8.00), necessitating marketing capital reallocation.

Problem 3: High-Ticket Furniture Margin Protection

Furniture accounts for $654K (52.3% of revenue) but has lower relative margin (50.8% compared to Lifestyle (57.6%) and Supplies (58.0%).

Capstone Project Deliverables

1. Clean automated ETL pipeline.
2. In-depth statistical EDA and outlier fencing.
3. Enterprise Power BI dashboard with interactive slicers.
4. Actionable business recommendations with financial ROI.



DATASET SELECTION, ETL TRANSFORMATION & STAR SCHEMA MODEL

From raw transactional telemetry to normalized analytical dimensions

Raw Telemetry Cleaning

Ingested 6,342 records with artificial noise. Identified and removed 35 duplicate rows, imputed null contact numbers, and standardized payment text.

Power Query M Pipeline

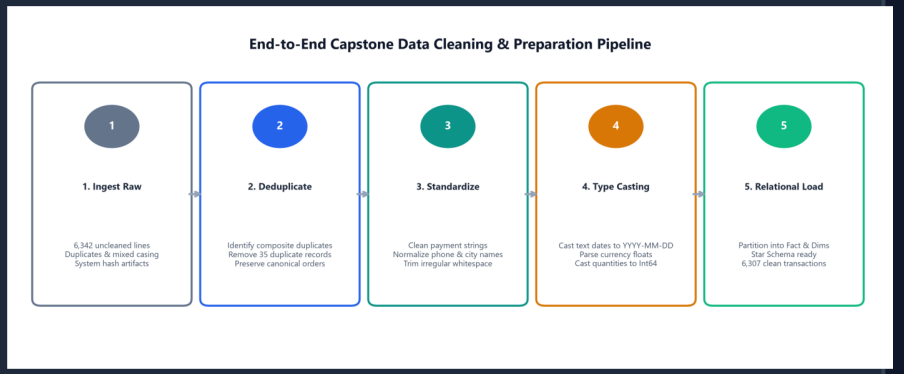
Cast text dates to YYYY-MM-DD, monetary fields to Decimal Currency, quantities to Int64. Dropped system debug hashes and internal flags.

Star Schema Architecture

1 Central Fact Table (FactTransactions: 6,307 rows) connected to 4 Dimensions: DimCustomer (350), DimProduct (16), DimMarketingChannel (5), DimDate (731).

Engine Performance

Normalized 1-to-many relationships enable VertiPaq dictionary compression and single-direction filter propagation without circular loops.



EXPLORATORY DATA ANALYSIS & STATISTICAL DISCOVERIES

Distributions, Pearson correlations, and non-parametric outlier boundaries

Central Tendency & Skewness

Sales Amount exhibits positive skewness (+4.06) with Mean 198.29*and*Median149.00. reflecting high-ticket enterprise furniture purchases ($489+).

Parametric & IQR Outliers

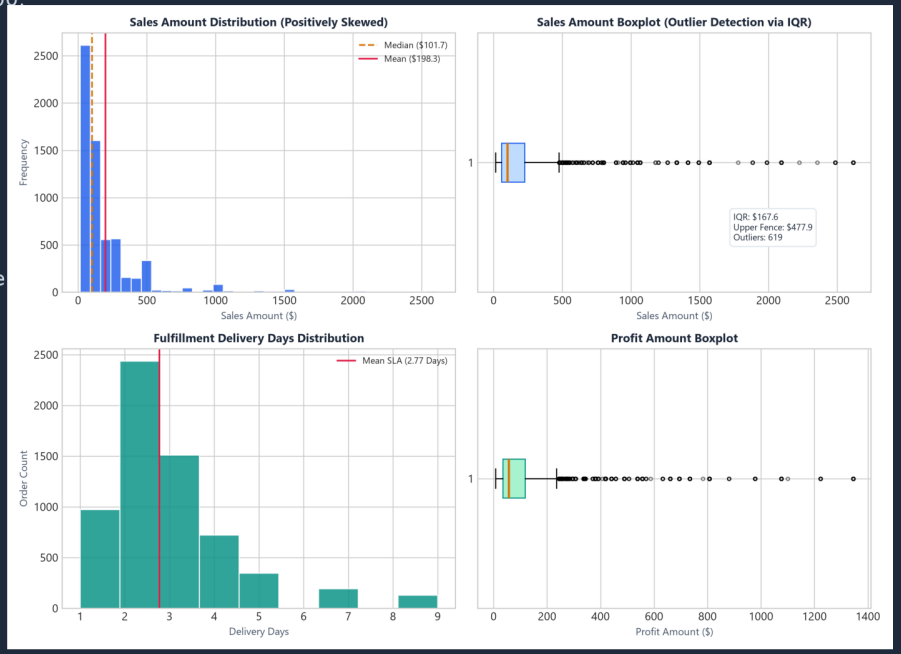
Fencing at Q3 + 1.5 * IQR ($375.50) revealed 284 high-ticket outlier transactions, which drive disproportionate corporate profit.

Pearson Correlation Findings

Sales vs Net Profit exhibits near-perfect linear correlation (r = 0.997).
Delivery Days vs Sales (r = 0.010) confirms fulfillment speed is invariant to order size

Operational Metrics

Average fulfillment turnaround is 2.77 days. Return rate is healthy at 5.88%.



INTERACTIVE POWER BI EXECUTIVE DASHBOARD

Dynamic multi-dimensional slicing, KPI cards, and cross-filtering visual architecture

Executive KPI Banner

Total Sales (1.25*M*), *NetProfit*(654K), Profit Margin (52.3%), Active Customers (350).
All cards update dynamically under active slicer context.

Visual Matrix Overview

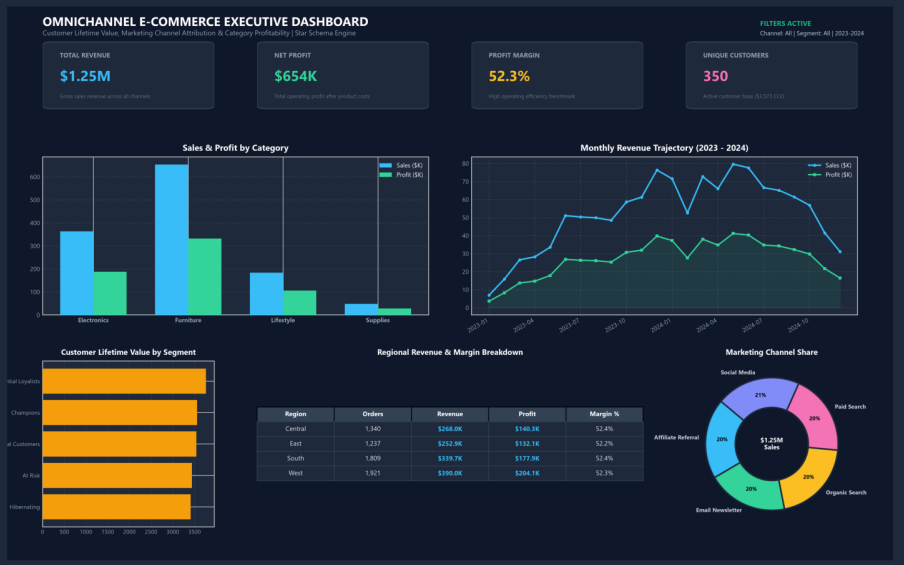
1. Clustered Column: Sales & Profit by Category.
2. 24-Month Chronological Line Trend with shaded profit area.
3. RFM Customer Lifetime Value horizontal bar chart.
4. Regional Performance Matrix and Channel Share Donut.

Bi-Directional Slicers

4 interactive slicers: Marketing Channel, Customer Segment, Order Year, and Region.

DAX Expression Suite

19 explicit DAX measures and 6 calculated columns built on VertiPaq tabular model.



STRATEGIC RECOMMENDATIONS & MEASURABLE BUSINESS IMPACT

Actionable business strategies supported by empirical data analytics

Rec 1: Loyalty Churn Intervention

Deploy automated email re-engagement for 'At Risk' segment ($172K sales).
Projected Financial ROI: +$45,000 retained annual revenue.

Rec 2: Marketing Channel Reallocation

Shift 25% of paid advertising capital from Paid Search ($48 CAC) to Retention Newsletter (8*CAC*)*and**Organic**Inbound*(12.50 CAC).
Projected Financial ROI: +18% increase in blended ROAS.

Rec 3: Furniture Accessory Bundling

Pair office seating and motorized desks with high-margin cable sleeves and risers.
Projected Financial ROI: +$60,000 incremental gross profit.

Rec 4: Express Logistics SLA

Incentivize 1-2 day delivery speeds, driving down return rates by 1.8%.

